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RESEARCH ARTICLE

Le Corbusier's urbanism: An urban characterisation of his proposals for inner cities

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Abstract Le Corbusier's urbanism is routinely met with a backlash of criticism, which often overlooks the complexity of his work. There is a tendency to focus on the characteristics of his proposal for Paris in 1925 or on the ideas of CIAM's Athens Charter. However, such affirmations ignore the conceptual and qualitative evolution of his urban proposals. In this sense, the research hypothesis is that Corbusian urban models go beyond the Plan Voisin, diffused by urban planning theories as the unique paradigm of his urban ideas.

Through analysis and characterisation of the inner-city urban plans of Le Corbusier, this report seeks to demonstrate the existence of other models and compares them to those often showcased in the partial examination of Le Corbusier's urbanism. The following parameters will be analysed: (1) population densities and areas for (2) buildings, (3) vehicles, and (4) pedestrians.

The results of this research indicate the need to parameterise and analyse Le Corbusier's cities beyond a single case to demonstrate the diversity and evolution of his urbanistic work. This process seeks to dissipate the common view that falsely extrapolates ideas of his urban planning, largely that of his first proposal, for Paris.
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1. Introduction

Le Corbusier's work and the man himself (1887—1965) are internationally recognised, and the impact he made during his lifetime continues to have an influence on the present day. As a result of his incursion into a variety of fields and subject matter, research into his proposals remains relevant and open to reinterpretation.

In this respect, the aforementioned international importance and influence of his work that occurred during the twentieth century have been acknowledged by the inclusion of at least part of his work on the List of UNESCO World Heritage Sites (2016). A total of 17 architectural works became a part of the list in 2016, and the preservation of these constructions became of international interest and made them the heritage of society as a whole. Among the selected works, some are located in the Chandigarh Capitol Complex (India), the only city of Le Corbusier to be built and a case study in this research.

Le Corbusier's architectural work is among his main achievements. Nevertheless, the diversity of disciplines undertaken by him goes beyond architecture, and an example of this is grouped under the concept of "synthesis of the arts" (Calatrava et al., 2020: 8).

In addition to amplifying his reputation as an architect, he enjoyed production on a grand scale as an urban planner, with a repertoire of more than twenty projects; and as a plastic artist. Paintings using different techniques, sculptures, and tapestries were the worthiest of note. He also created furniture, in addition to being a writer and theoretician of material that spans almost his entire career. A large proportion has been collected and is now in the *Fondation Le Corbusier* in Paris.

Le Corbusier was such an interdisciplinary figure that a number of different aspects are combined in his works. One side is sometimes inseparable from the others. Therefore, regarding urban planning, it is crucial to understand the urban-theoretical formula that characterises him: "It is not possible to separate the texts that Le Corbusier dedicated to urbanism from that of the rest of his professional career" (Barbarin and Luque, 2004: 555).

When investigating the origins of the Fondation Le Corbusier, it could be inferred that the architect-urban planner did not want to leave anything to chance. Therefore, before his passing, Le Corbusier could already be found designing and laying the groundwork for a site in which the materials produced throughout his career would reside, as implied by a text summarising the history of the foundation (Fondation Le Corbusier, n.d.). Currently, it is the most important source of information for his work and the place researchers go in order to pick apart the intricacies surrounding each of his works. Consequently, the Fondation Le Corbusier constitutes an essential part of this research. Its digital archive lends itself to the consultation of image documentation for each city under analysis.

Among all the disciplines undertaken by the Swiss-French master, this research is focused on urban planning. The general assumption is that Le Corbusier's urban contributions were based on a series of parameters and characteristics repeated in each proposal throughout his career. This assumption is invariably supported by past and present publications, focusing primarily on the 1925 Plan Voisin for Paris (Colmenares, 2019: 67; Kunzig, 2019: 27; Sennett, 2019). The recurrent use of this proposal results from its contribution to modern urban planning and the concepts taken from the Athens Charter in 1933 (Le Corbusier, 1943).

A tendency to address his urban proposals via only a small selection of his works has been a constant in architecture reference books worldwide, old and new, in which urban issues are covered at large. Publications such as *Historia crítica de la arquitectura moderna* (Frampton, 2009) and *Historia de la Arquitectura Moderna* (Zevi, 1954) tend to focus on his designs for La Ville Contemporaine de trois millions d'habitants and La Ville Radieuse, as well as realised projects such as the aforementioned Plan Voisin and that of Chandigarh. The descriptions given are only brief.

In this regard, Kenneth Frampton's position on the matter is worth highlighting: he acknowledges "the other Le Corbusier", that is, looking beyond his initial link to purism-mechanisation, which he cut ties with at the beginning of the 1930s (Frampton, 1987: 30). Richard Sennett points out the evolution of Le Corbusier's urban projects in similar terms. He highlighted the notable differences between the Plan Voisin and Chandigarh, which transcended "Le Corbusier's earliest convictions and lack of experience" (Sennett, 2019: 104).

Given this rationale and light of the incomplete interpretations of his work and the vagueness surrounding its context, there is a pressing need to address the projects and other such things that he undertook beyond his first years. In doing so, there is an advocation for overcoming and breaking away from preconceived ideas surrounding Le Corbusier (Sequeira, 2013: 5).

Although the Plan Voisin and Chandigarh are the works addressed most often in the literature consulted, it should be noted that several of the cities proposed by Le Corbusier have undergone detailed analysis and been thoroughly researched. Indeed, some of them appear in publications by MoMa (Cohen (Coord.), 2013) and in the Spanish magazine *AV Monografías 176* (Fernández-Galiano (Dir.), 2015). In addition, it would appear that his Bogota proposal has experienced a remarkable level of analysis, namely, in the case of María Cecilia O'Byrne's research (2008, 2018) and in that of Doris (Tarchópulos Sierra, 2010).

In terms of cross-sectional readings as a whole, Xavier Monteys' doctoral thesis and subsequent publication *La Gran Máquina. La Ciudad en Le Corbusier* (1996) stands out for already conducting such research and has been an essential bibliographical source for developing this research. However, its focus is mainly historiographic. It is written in an essay style and is descriptive in nature, as acknowledged by the author in the synopsis.

Therefore, the research hypothesis is that Corbusian urban models go beyond the Plan Voisin, diffused by urban planning theories as the unique paradigm of his urban ideas.

2. Objectives

The partial insight into his work, based on the examination of only one of more than twenty urban projects developed over the course of his career, contributes to an incomplete understanding of Le Corbusier's urbanism. Moreover, it is usually paired with a lack of knowledge regarding a contextual analysis or an objective characterisation using urban parameters. The need to conduct in-depth studies to better understand his work's vicissitudes is all too often ignored. Consequently, this research does nothing more than perpetuate preconceived ideas.

Considering the importance of analysing the context in which the urban planning proposals were made, it is regarded as equally important to address his work as a whole where possible. Therefore, this proposal conducts holistic natured research to demonstrate the significant evolution experienced by Le Corbusier (Rodríguez-Lora et al., 2020).

Thus, the main objective of the research is to characterise and parametrise a group of Le Corbusier's cities to highlight the diversity of his work. The study sample is composed of Le Corbusier's urban planning proposals featuring inner cities (Table 1, Fig. 1, Fig. 2). In contrast to those located on the coast, the various urban dynamics of these cities were crucial to creating the sample analysis. They also allow us to include his first proposal for Paris and Chandigarh, his only build.

Furthermore, the proposed parameterisation seeks to demonstrate the need for broader analyses of his work, to highlight the conceptual evolution of his urban planning over time, and to try to overcome preconceived ideas about Corbusian urbanism.

3. Research method

This research was structured around three consecutive and interrelated phases. Therefore, the methodology focused in particular on obtaining data intended for the characterisation and parameterisation of the cities in the study.

Therefore, the variables selected for the analysis are as follows: demography (inhabitants and density), areas for use (residential/other), built-up areas, and areas for mobility (pedestrianized areas/footpaths/roads).

First, a bibliographic and planimetric search was conducted in order to generate the database used in the research. Once the group of cities that composed the study sample was determined, it was then necessary to obtain a new bibliography and planimeter for each of them for their future characterisation.

In order to proceed with the task at hand, an examination of the research and publications by various authors was conducted; and the bibliographical sources and documents considered essential or of the highest interest, with a focus on the study of Le Corbusier's urban planning profile, were in turn selected and examined. The previously mentioned *La Gran Máquina. La ciudad en Le Corbusier* by Xavier Monteys (1996) stands out in particular for its combination of all of the architect's urban projects in different stages.

Likewise, numerical data of the *Plan Director de Bogotá* (LC08) were obtained consulting the plan's records, recovered from *Le Corbusier in Bogotá, 1947—1951. Anexos* (O'Byrne Orozco (Coord.), 2008).

The consultation of documents produced by Le Corbusier himself was indispensable. Data were obtained from three of the books from the *Le Corbusier et Pierre Jeanneret. Oeuvre Complète works —1938-1946* (Boesiger (Ed.), 1995), *1946—1952* (Boesiger (Ed.), 1995a) and *1952—1957* (Boesiger (Ed.), 1995b) — as well as from the publication *La Ville Radieuse* (Le Corbusier, 1964).

Equally, the *Fondation Le Corbusier* digital archive, which has allowed downloading the designs from the architect's studio, proved another indispensable component in the process. Using this tool, incomplete areas of the numerical data obtained from the bibliography could be completed planimetrically using graphical estimations.

In the second phase, the planimeters of each city were redrawn using CAD in order to obtain more data. Once the information necessary to complete the database in the first phase had been identified, in order to be able to obtain data on surface areas, the selected cities were redrawn based on the planimeters acquired from the Xavier Monteys' document (1996) and the *Fondation Le Corbusier*.

The software was used to scale each of the plans and to draw each of their elements so that the surface areas could be calculated. The calculations drawn from specific areas allowed the surface areas for each of the uses to be obtained.

Finally, in the third phase, graphs were developed to enable cross-referencing. Thus, through graphical representation of the numerical data recorded throughout the research, the similarities and divergences present throughout the entire analysed sample are demonstrated.

Therefore, the decision was made to use linear graphs set over a timeline that shows the trajectory and characteristics of the cities. The clear display of the varying cities in this particular representation allows for cross-sectional examination that sheds light on the complexity of Le Corbusier's urban development beyond examining the plans in isolation.

Table 1 Selection of cities for the study. Source: own production.

Code	Year	City	Country
LC01	1925	Paris	France
LC02	1933	Geneva	Switzerland
LC03	1933	Antwerp	Belgium
LC04	1935	Zlin	Czech Republic
LC05	1935	Hellocourt	France
LC06	1945	Saint-Dié	France
LC07	1945	Saint-Gaudens	France
LC08	1950	Bogota	Colombia
LC09	1950	Chandigarh	India
LC10	1955	Meaux	France
LC11	1958	Berlin	Germany

4. Findings

4.1. Inhabitants in Le Corbusier's cities

Le Corbusier worked on multiple urban scales, not only in terms of surface area but also in terms of population numbers to be considered in each case. This alone gives an idea of the divergent magnitude of the study sample, which ranges from projects for capital cities to others of minor importance or on a much smaller scale. It also includes limited contributions made within larger cities, as in the case of Geneva (LC02). The range of inhabitants (Fig. 3) varies from 5000 for Saint-Gaudens (LC07) to 500,000 in the case of Antwerp (LC03).

Cases such as Zlin (LC04), which has the second-largest population in the sample at 290,000, and other groups slightly below this, at approximately 150,000—200,000, which include Berlin (LC11) and the first phase of Chandigarh (LC09), are between these two ranges. The Plan Voisin for Paris (LC01), with approximately 78,000 inhabitants, and the portion analysed from Bogota (LC08), with 64,000 inhabitants, are in intermediate positions. Below that are Hellocourt (LC05) and Meaux (LC10), with slightly above 30,000 inhabitants. Finally, close to the minimum are Saint-

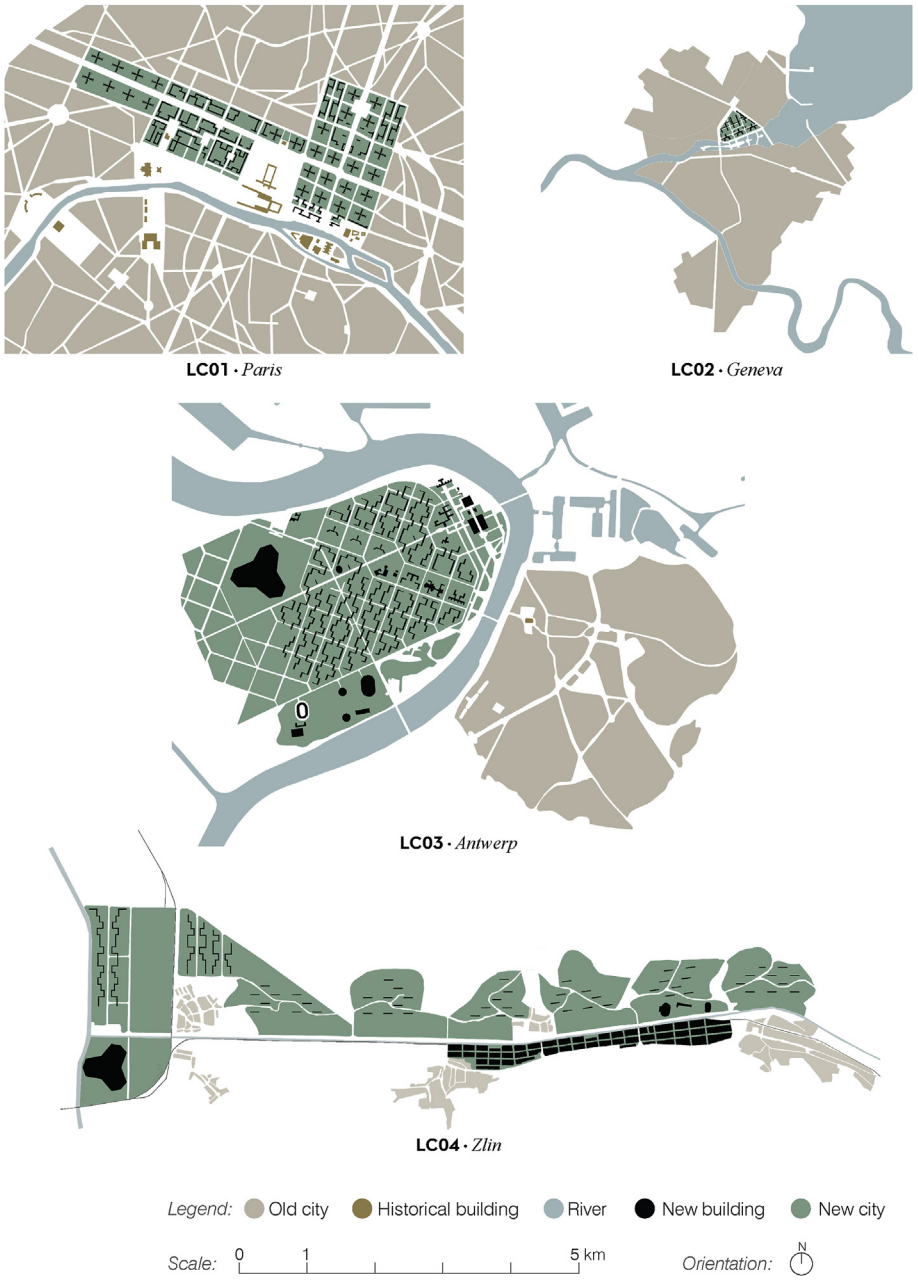


Fig. 1 Representation of the cities in the study on the same scale (LC01 to LC04). Source: own production.

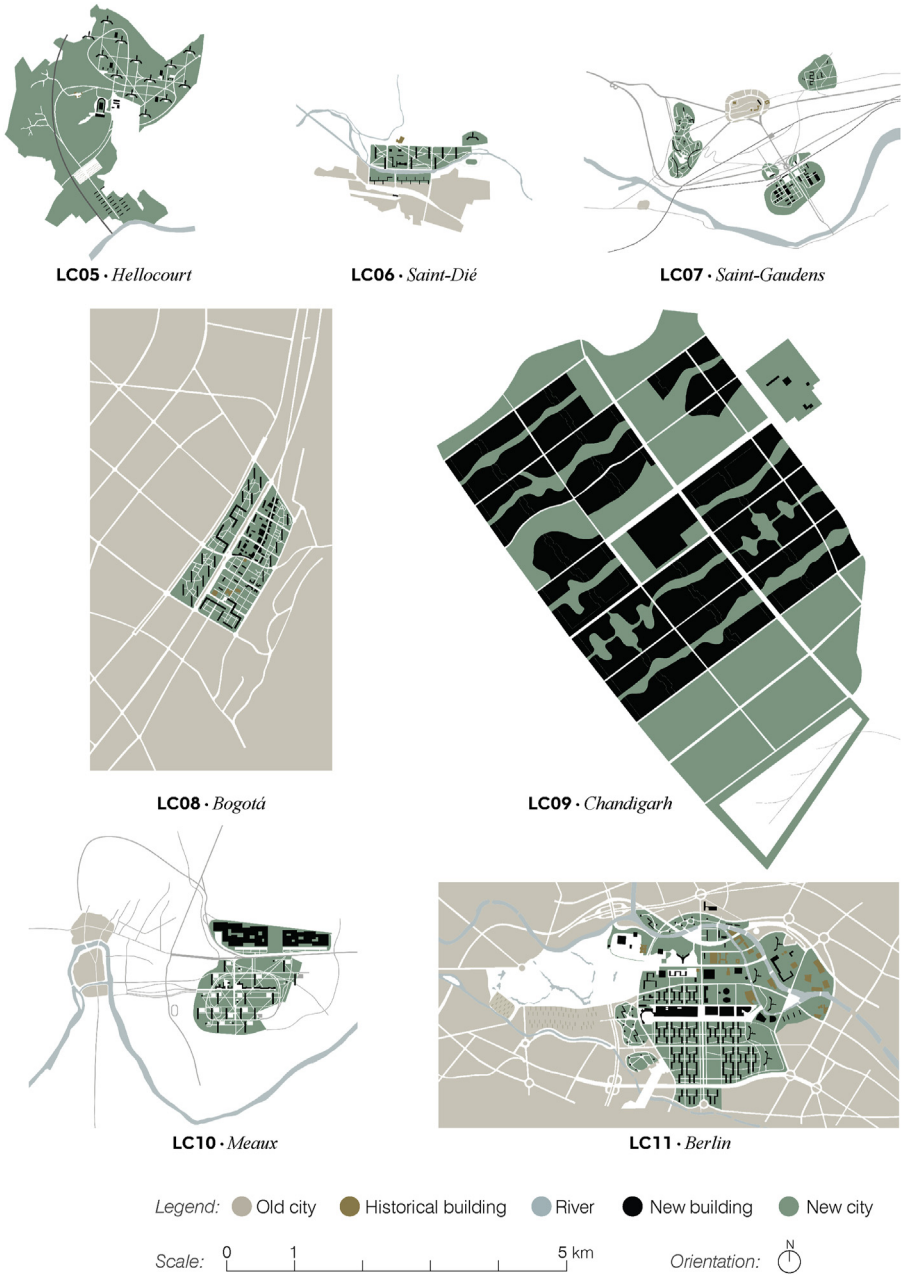


Fig. 2 Representation of the cities in the study on the same scale (LC05 to LC11). Source: own production.

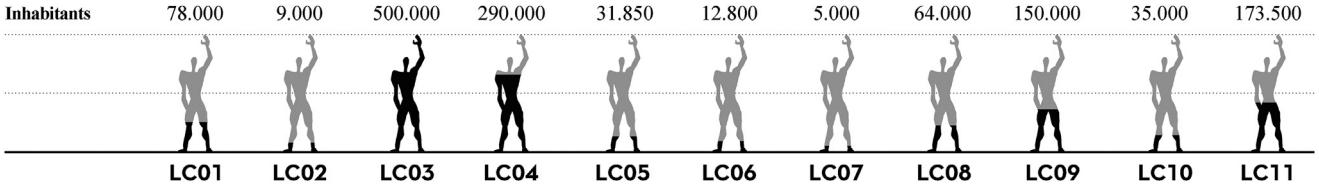


Fig. 3 Inhabitants of the cities analysed. Source: own production.

Gaudens (LC07), Saint-Dié (LC06), with approximately 12,800 inhabitants and the analysed portion of Geneva (LC02) with 9000.

To relate the population data with the surface area of the proposals, population densities were calculated and estimated in terms of inhabitants per hectare (Fig. 4).

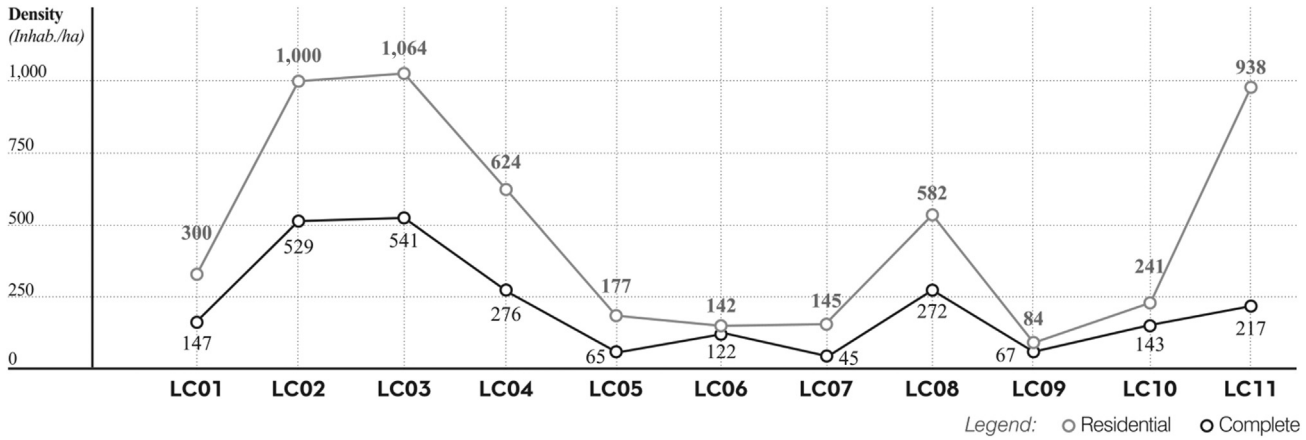


Fig. 4 Densities of the cities analysed (inhab./ha). Source: own production.

It should also be noted that several of the density figures were obtained with the aid of graphic estimation according to the type of building, the surface area allocated per inhabitant by Le Corbusier in each phase, and/or data sourced from the corresponding bibliography. Despite being composed of different sources, this case intends to review the evolution of the urban parameter of density throughout its trajectory. Finally, it is essential to note that research considers the differentiation of densities according to whether the cases are exclusively residential areas or the entire proposal, regardless of the general use. This is represented in two levels or layers of densities for each graph and analysis (Fig. 4): residential (grey) and complete (black).

After a thorough reading of the documentation and books written by Le Corbusier himself, it is clear that one of his strategies when planning the modern city was densification, aiming to achieve it through building height. This was the case in his earliest projects and made more evident by his proposals for the Ville Contemporaine, his plan for Paris and the Ville Radieuse.

This commitment to densification was a reinterpretation of the garden city. That is, he considered a greater integration of nature and vegetation alongside the built environment to be the ideal: "Live! (Breathing) We must do away with the suburbs and bring nature into the cities" (Le Corbusier, 1964: 104). However, he rejected the idea of placement in the periphery and low density because of the land consumption it required and an increase in displacement. Proposals for high buildings with a free floor plan for pedestrians and the buildings integrated in the city centre instead of in the periphery emerged from this analysis.

Thus, the Plan Voisin (LC01) serves as a clear example of developing such concepts in their initial stages. Its characterisation demonstrates that the proposed number of inhabitants was approximately 300 inhab./ha for the predominantly residential areas, and it was almost half that for the complete proposal: 147 inhab./ha.

The redensification proposals are especially visible in his design for the Ville Radieuse, which proposes a higher density level than was usually seen in city centres, doubling it even: "Thousand per hectare instead of 400—500" (Le Corbusier, 1964: 110). Thus, a substantial increase can be seen in the first two cities where this design is applied to a greater extent, Geneva (LC02) and Antwerp (LC03), reaching 1000 inhab./ha, according to the book *La Ville Radieuse*. This increase compared with the rest of the sample can also be identified in the Ville Verte plan, where the *redent* layout is proposed.

It should be noted that among the Corbusian proposals analysed, this kind of density was not repeated until his last project, which was submitted in the competition for Berlin (LC11), where he revived levels of densities similar to those proposed for the Ville Radieuse.

A decrease in density can be seen in the proposal for Zlin in 1935 (LC04), being approximately coetaneous to Geneva (1932) and Antwerp (1933). Nevertheless, it remained at a higher level than that of Paris, in this case, duplicating his initial proposal.

In Zlin, where a design change was made, more akin to the linear city, Le Corbusier chose a density of 624 inhab./ha for residential areas and 276 inhab./ha for the complete area, showing the duplication of the French city, but with levels of approximately 62% of the immediately preceding plans.

The downward trend is strengthened in the following three cities analysed, Hellocourt (LC05), Saint-Dié (LC06) and Saint-Gaudens (LC07), which show populations on a medium-low scale. This casuistry can be observed in three cities where three different models are applied (Ville Radieuse, CIAM, and Ciudad Lineal Industrial), with a decade separating the Hellocourt proposal from that of Saint-Gaudens. The latter has the lowest density of the sample for the complete urban area, 45 inhab./ha; and the second lowest values of this subgroup with 145 inhab./ha. The density values for the other two cities are as follows: Saint-Dié shows densities of 142 inhab./ha (the lowest value of the compared trio) for the residential area and 122 inhab./ha for the complete area. Hellocourt has a density of 177 inhab./ha for the residential area and 65 inhab./ha for the entire zone.

However, this trend increases in the Bogota Master Plan (LC08), the first city in the sample where the 7 V model is applied, increasing the density to the levels in the proposal

for Zlin (LC04). The data for this city stand at 582 inhab./ha for the residential area and 272 inhab./ha for the complete area proposed.

A comparison between the densities proposed for Bogota (LC08) and Chandigarh (LC09) shows the adaptability of the models, even in projects conducted almost simultaneously (around the 1950s) and having the intention of catering to two very different contexts, America and Asia. However, both cities enjoyed high levels of institutional representation due to their capital status.

Regarding Chandigarh (LC09), there is a return to the density levels that characterise the downward trend shown from the mid-1930s onwards. It is worth noting the similarity in densities, both in residential areas and when considering the rest of the city. This convergence demonstrates a greater variety of uses intended in the case of the Indian city, where the data almost coincide. It includes 84 inhab./ha for the residential area versus 67 inhab./ha for the complete area. It should also be noted that these are the lowest numbers for the entire sample. The uniqueness of Chandigarh is evident in several ways, including the components and scale of the city, which is lower in height than any of the other cities analysed.

Last, in the final stretch of his urban planning, there is a significant increase in density, especially in Berlin (LC11). The density of Meaux (LC10) reaches levels close to those of Paris, with 241 inhab./ha in the residential area and 143 inhab./ha in the complete area. These data show an increase in, and rethinking of, the density in the modern city. In the case of Berlin, densities close to that of the Ville Radieuse are seen, approaching 1000 inhab./ha. Based on the estimations, a density of 938 inhab./ha in the residential area and 217 inhab./ha in the complete area was calculated, considering only the residential area or the entire city.

4.2. Surface areas

As part of the characterisation of Le Corbusier's urban projects, an analysis was made of the surface areas and their intended use in each of the proposals, highlighting the importance placed on predominantly residential use in the Corbusian city. The use of the surface area for nonresidential and tertiary buildings and elements appears on a lesser scale, although both urban functions are compatible. Nonresidential and tertiary uses will be referred to collectively as "others".

Likewise, the total percentages of built-up surface area in relation to open space, whether having been allocated to roadways or recreational areas, are also demonstrated. Next, all of the space not built upon was analysed, and the results showed how much of it was allocated to pedestrian movement and vehicle movement.

The final analysis focuses on defining how much of the pedestrian space is concentrated on paths and walkways. The aim is to demonstrate how the networks of paths play a greater or lesser role in each city, which cities feature a network for pedestrian movement, and how cities' pedestrian routes have the most negligible impact on open spaces.

Last, it should be noted that all of the surface areas in question were obtained via graphic analysis and measurement.

The first analysis (Fig. 5), which focuses on the total surface areas in use, shows that there is a tendency towards equal distribution, approximately 50% for residential use and similar in other uses. This tendency is especially clear in Le Corbusier's initial urban proposals. Therefore, it can be seen in cities such as Paris (LC01), Geneva (LC02), Antwerp (LC03) and Zlin (LC04). Such tendencies and overall balance are only seen again in the case of Bogota (LC08), with 53.2% of the use allocated to "others".

The percentage allocated to other uses besides residential rises to 63.3%, going slightly higher than the 50% mark commonly seen in previous projects. A similar range, close to 60%, is seen in later proposals, such as Saint-Gaudens (LC07) with 68.6% allocated to "other", Hellocourt with 63.3%, and Meaux (LC10) with 59.2%, though it is allocated to residential use. Berlin (LC11) has the highest value, with 76.9%.

Besides, the greatest imbalances with respect to the indicated trend are found in Saint-Dié (LC06), where 85.7% is allocated to residential use compared to an area of 14.3% for other uses; and in Chandigarh (LC09), with 79.1% and 20.9%, respectively.

The second analysis focuses on the proportion of built elements in the form of buildings in relation to open space, consisting of roadways, areas allocated for pedestrian activity and green space.

Ten cities in the sample have much lower percentages of built-up area than unbuilt area (Fig. 6), and four have an intermediate percentage of approximately 10% of built-up area. Below these are another four, the lowest being Hellocourt (LC05) at 2.6% built-up area. The highest of the ten are Meaux (LC10) and Geneva (LC02), which have the aforementioned 22% of built-up area.

The only city proposal in which this does not occur is Chandigarh, where built-up areas amount to 63.2% of the total area proposed.

Having characterised the presence of the open space in the Corbusian city, it is necessary to further analyse it by parameterising the allocated uses for its total surface area. Thus, as a third analysis in this section, the percentage of each city allocated exclusively for pedestrian use and road use is shown (Fig. 7). The importance of roads for the modern city is expressed by Le Corbusier himself in his publication of *El urbanismo de los tres establecimientos humanos* (Le Corbusier, 1964a: 37).

The graph (Fig. 7) shows that the lowest percentage of the study sample is 9.5% in the case of Saint-Dié (LC06). Four other cities have approximately the same proportion: Antwerp (LC03) and Berlin (LC11), with 11% and 11.1%, respectively; Saint-Gaudens (LC07) with 12.2%; and Meaux (LC10), with the highest in the grouping, with 15.5% of its surface area allocated to road use.

At double the lowest figure is the plan for Zlin (LC04), with 20.3%; and Bogota (LC08) with 22.1%. The first two

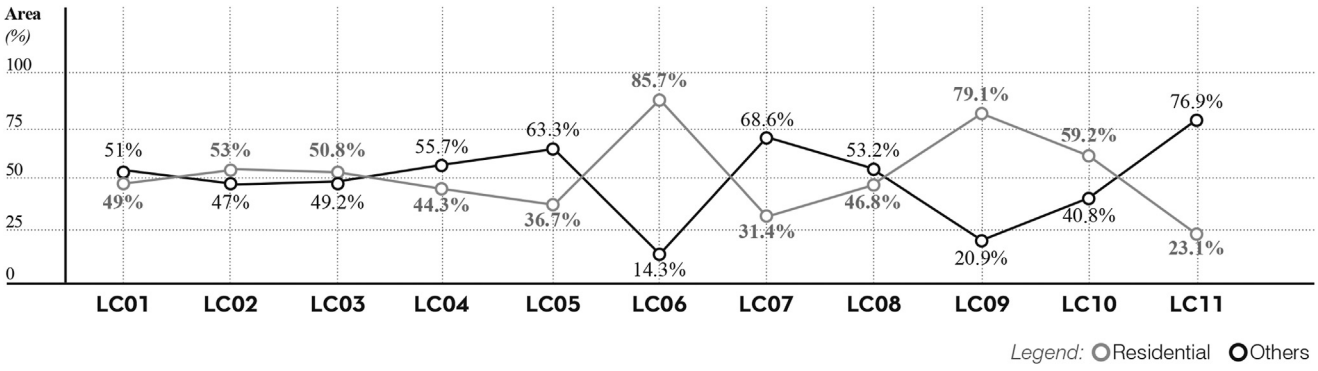


Fig. 5 Percentage of surface areas allocated to residential use compared to other uses. Source: own production.

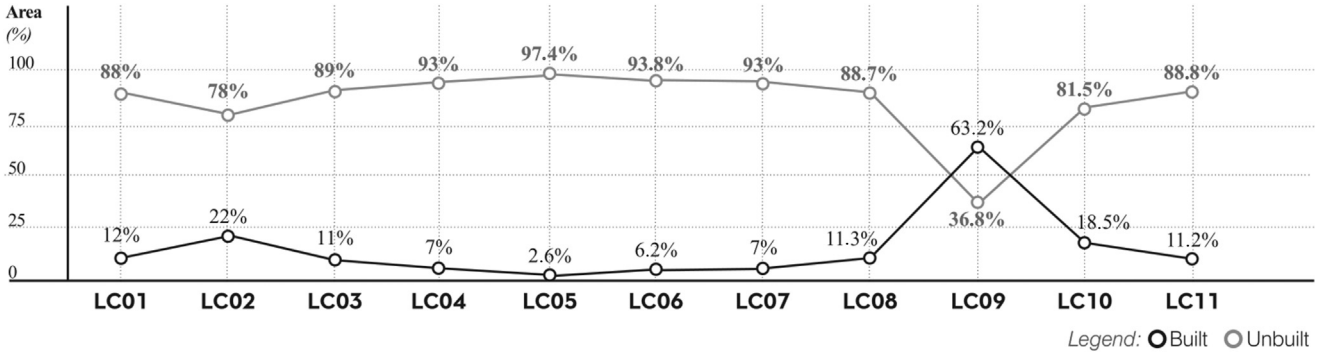


Fig. 6 Percentage of built-up areas compared to open space. Source: own production.

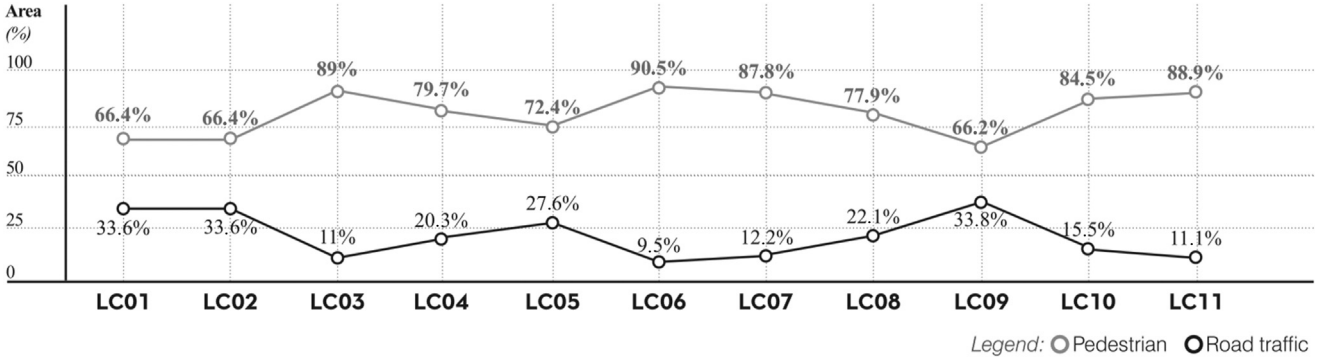


Fig. 7 Percentage of areas allocated to pedestrian and road use. Source: own production.

urban proposals analysed in this research are those that have the highest percentages. Both Paris (LC01) and Geneva (LC02) allocate 33.6% of their surface area to vehicle movement, highs which are not seen again until the proposal for Hellocourt (LC05), despite the roadway being a simplified closed circuit in the shape of an "8" with two linear bypasses leading away from it. The Chandigarh master plan (LC09) allocates the largest area to vehicle movement, with 33.8% of the open space allocated to it after deduction of all built-up areas.

In terms of areas marked as being free of vehicle circulation, two uses of the space have been identified: one as a space for recreation and the other as a network of paths and pedestrian routes overlapping the road network. The importance of these urban walkways is not always equally considered, with them taking precedence in some cities more than others, as detailed below.

It should be noted that, in the most extreme cases, potentially that of Paris (LC01) where paths are not proposed, there was a slight trend towards doing so after analysis of the Plan Voisin design. Although the network does not have a sufficient scope to be able to be defined as a pedestrian movement system, it could be that the movement is less guided than in other cases. In Chandigarh (LC09), however, the opposite is true. The large spaces are taken to be pathways because of the direction they take.

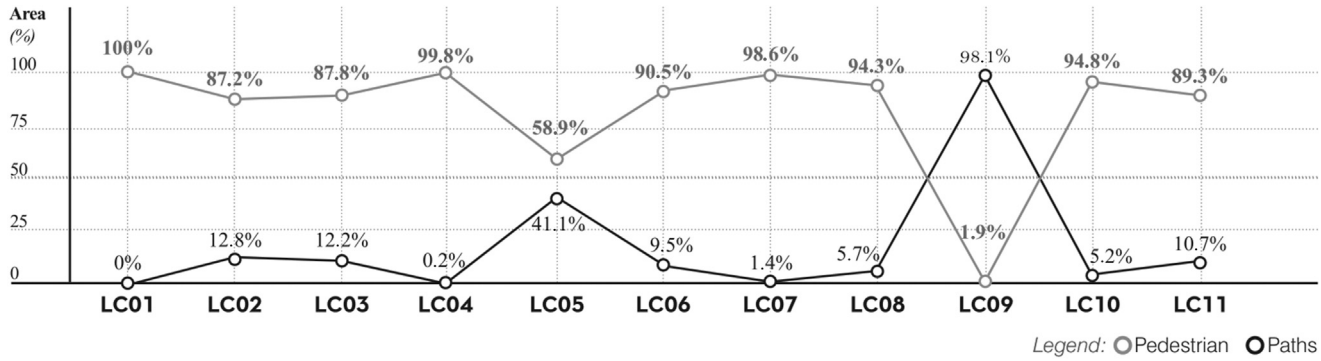


Fig. 8 Percentages of surface areas allocated to generic pedestrian areas and to the network of paths and routes for pedestrian movement. Source: own production.

Without losing sight of the changes in surface area that the public space undergoes, it manages to accommodate recreational spaces while at the same time directing pedestrian movement.

The analysis of the obtained parameterisation data of their characteristics shows the evolution of this pedestrian network's importance throughout Le Corbusier's active career as an urban planner (Fig. 8).

Thus, starting from the 0% in Paris, Zlin (LC04) with 0.2% and Saint-Gaudens (LC07) with 1.4% are the cases that present the lowest percentages of his surface allocated to paths. These last two cities share the common factor that they are based on the Ciudad Lineal industrial model and that, together with the third in the sample based on this model, Meaux (LC10) with 5.2%, they emerge as those in which a pedestrian network is least important, alongside the aforementioned plan for Paris. Bogota should also be added to the list (LC08) at 5.7%.

A little above those previously mentioned cities are Geneva (LC02) at 12.8%, Antwerp (LC03) at 12.2%, Saint-Dié (LC06) at 9.5%, and Berlin (LC11) at 10.7%. Their plans also have common factors in that they are all based on the same urban model. Even though paths appearing as a network are displayed clearly and with unquestionable prominence in the plans for the Ville Radieuse model, and especially in those of the Ville Verte, in the first two plans mentioned, this was not the case. This case contrasts with Hellocourt (LC05), which is also based on this model and had the second largest surface area of paths, 41.1% in total, in the sample studied. Saint-Dié and Berlin, both designed under CIAM concepts, have approximately 10% of their surface allocated to paths.

Finally, Chandigarh (LC09) stands out at the top of the sample, with 98.1%. It is the only urban project to exceed the 50% threshold.

Regarding the total area allocated to public space per inhabitant, as the final analysis of this section, variations were noted across the studied trajectory, with cases that have values substantially exceeding those of others (Fig. 9).

The lowest values are those seen in Geneva (LC02) and Antwerp (LC03), with 14.8 m^2/inhab. and 16.5 m^2/inhab., respectively. This similarity is reinforced by the proximity in time between both proposals—1932 and 1933—and by both having been based on the same Ville Radieuse model. This is not the case for the other city based on the same model and developed a few years later, Hellocourt (LC05), with a profile closer to that of the "industrial green city" developed by Le Corbusier. The differences between the latter and the previous cases are substantial. Hellocourt allocates 111.5 m^2/inhab., making it the city with the second-largest surface area per inhabitant allocated for pedestrians.

The higher proportions are claimed by Zlin (LC04) with 28.9 m^2/inhab., Bogota (LC08) with 28.6 m^2/inhab., and Chandigarh (LC09) with 35.4 m^2/inhab., despite being among the cities with the most space allocated to vehicles as opposed to pedestrians.

The Plan Voisin is characterised by its proposal of a surface area of 45.1 m^2/inhab., a figure that is not repeated until the end of Le Corbusier's career as an urban

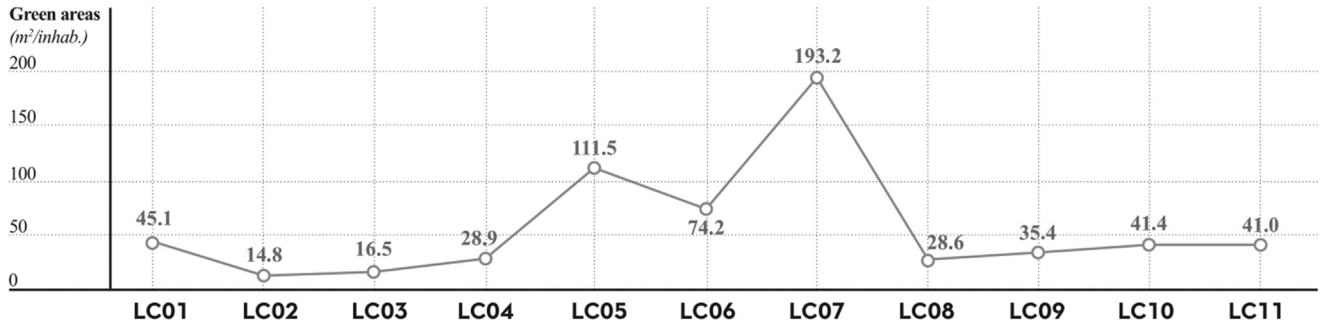


Fig. 9 Public space (m2) per inhabitant in each city. Source: own production.

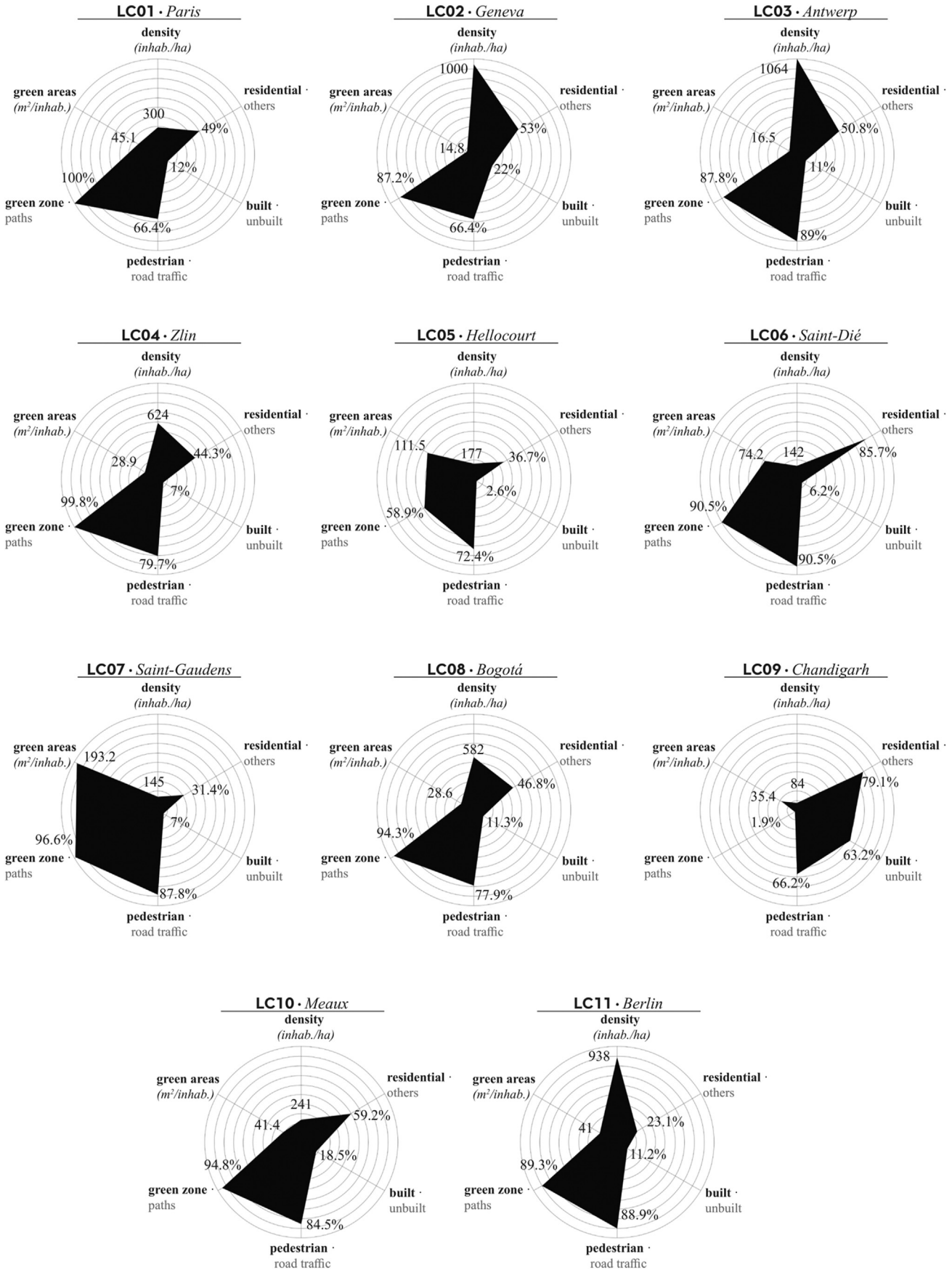


Fig. 10 Synthesis image of all the results. The data are from the word in bold. Source: own production.

planner when the project for Meaux (LC10) had 41.1 m^2/inhab. and finally that of Berlin (LC11) had 41.0 m^2/inhab. In this case, there is no evident relation between models. However, they are close to 40 m^2/inhab., a trend that started to develop following the plans for Bogota and Chandigarh.

The central proposals of the studied sample stand out in terms of the largest surface areas allocated to this purpose. In addition to Hellocourt, which was previously mentioned, Saint-Dié (LC06) is notable in third place, with 74.2 m^2/inhab.; and, with a surface area much larger than the rest, Saint-Gaudens (LC07) with 193.2 m^2/inhab.

The joint representation of all the data from the characterisation shows different common patterns for their urban projects (Fig. 10). Although each project has particularities, his urban proposals have been classified into five different groups. First, the initial case of Paris (LC01) can be associated with one of the last cases, Meaux (LC10). Something similar would happen with two other initial cases, Geneva (LC02) and Antwerp (LC03); and the last case for Berlin (LC11). Two other groups can be identified and are at an intermediate stage of his career: Zlin (LC04) and Bogotá (LC08) and Hellocourt (LC05), Saint-Dié (LC06) and Saint-Gaudens (LC07). Finally, Chandigarh (LC09) appears as a case different from the rest.

5. Discussion

In terms of population, there are proposals between 500,000—290,000 inhabitants (2 plans: Antwerp and Zlin), 200,000—150,000 inhabitants (2 plans: Chandigarh and Berlin), 78,000—64,000 inhabitants (2 plans: Paris and Bogotá), 35,000—30,000 inhabitants (2 plans: Hellocourt and Meaux), and 12,800—5000 inhabitants (3 plans: Geneva, Saint-Dié and Saint-Gaudens). Therefore, up to five levels of population size can be marked out, whereby out of all the analysed plans, two plans fit into each one of the ranges.

Concerning the proposed cities, this diversity demonstrates Le Corbusier's adaptability to even the most heterogeneous situations among the various theoretical designs he developed. The differences in scale and population between the smaller and the larger designs reflect the urban planner's intention to provide modern city solutions regardless of the city's characteristics by understanding urbanism not as something relative to quantities but instead focused on attaining an intended urban quality. When analysing the obtained density data, similarities between values can be used to group the proposals. Thus, considering all of the Le Corbusier studied proposals and using the density of inhabitants per hectare for residential areas as a singular unit, four levels can be defined.

With densities of approximately 1000 inhab./ha, Geneva, Antwerp and Berlin are at a high level. At an intermediate level, close to 500 inhab./ha, are Zlin and Bogota. Next, at a lower level within the sample are the cities with 150—250 inhab./ha, such as Hellocourt and Meaux. Finally, with fewer than 150 inhab./ha are Saint-Dié, Saint-Gaudens and Chandigarh.

Relations between the cities and the shape of the curve shown on the density data graphs demonstrate a similar trend, although with some nuances that develop over time, regardless of addressing residential areas or complete urban proposals. This fact reflects the diversity of densities proposed by Le Corbusier throughout his career beyond his first proposal for Paris.

At the beginning of its career, a trend can be identified focusing on analysing the areas assigned for each use: approximately 50% is used for residential and other functions, as in Paris, Geneva, Antwerp or Zlin. However, the project for Paris is particularly significant due to the cruciform tower areas. The height of the towers opposes an extensive surface area expansion where the *redent* buildings are located. Something similar occurs with Zlin, where the surface area allocated for industry is especially relevant.

Simultaneously, a similar case of Zlin occurs in Hellocourt due to the remarkable development of areas allocated for the industry. However, recreation and sport areas are also noteworthy in this particular case, such in the vicinity of lakes.

Berlin, Saint-Dié and Chandigarh have the most considerable imbalances. In these last two cases, the importance of housing is particularly evident; and as demonstrated in the case of Chandigarh, there is an outstanding commitment to using residential areas in multiple ways and minor specialisation of the urban areas. The analysis of his Master Plan design and the resulting built city can demonstrate the previous features. Berlin is the city that has the most significant weight of other uses, considering its characteristic of being a capital city.

The second analysis shows the percentages of surfaces built and unbuilt obtained across the study sample, demonstrating the predominance of open space in Le Corbusier's modern city compared to the built-up rates. In most cases, the built-up area is between 2.6% and 22%. Thus, the prominence of open space in the Corbusian city is demonstrated.

This does not happen in Chandigarh. In this case, rather than being a strictly urban model, the 7 V model is an adaptation of the concepts geared towards the particular context and casuistry of India. This feature has already been demonstrated in the density study, given that it has the lowest density for residential areas. Despite this, as illustrated in more detail below, the Chandigarh proposal does not rank the lowest in terms of green space per inhabitant.

The third analysis highlights that of all the urban plans studied, pedestrian areas are substantially larger than those used by vehicles and have been enlarged since then. In terms of mobility, the surface reserved for cars is significantly reduced in some case studies. This finding contradicts the generalised idea of vehicle predominance over pedestrians in the Corbuserian city.

Fourth, regarding the surface of pedestrian paths, although at the beginning Le Corbusier proposed a few unconnected routes, the pedestrian networks became an essential element in the cities he proposed. The width of the walkways varies. Thus, as mentioned above, Chandigarh is the city with the largest surface area allocated to pedestrian movement, without losing sight of the characteristics already indicated in the findings.

Finally, in terms of green area per inhabitant, the research found similarities between the year in which they were conceived and the models applied. This is the case for Geneva and Antwerp, which have the lowest values of the sample.

Next, there are cities such as Zlin, Bogotá and Chandigarh. While Zlin is based on the lineal industrial city model, the remaining two cities follow the 7 V model, demonstrating similarities in terms of the surface area per inhabitant analysed. Nonetheless, irrespective of the model, there is a trend towards the same values and back to the initial Paris level in the final stretch of his career as in Meaux and Berlin.

The cities with the highest levels of surface area per inhabitant are Hellocourt, Saint-Dié and Saint-Gaudens. As in the previous case, this is more of a temporary trend than a link to the model in question since, once again, there are three models in play: Ville Radieuse, CIAM and Ciudad Lineal industrial. The cities that uses the latter model are the cities that allocate the most open space per inhabitant.

Given the groups formed after representing all the parameters obtained in the characterisation, the diversity of proposals is evident despite certain similarities.

Although there are parallels between the first proposals and the last proposals, such as Paris, Geneva and Antwerp with Meaux and Berlin, the central core of the proposals shows a clear differentiation. In this sense, the cities of Hellocourt, Saint-Dié and Saint-Gaudens, which are some of the smaller scale cities in the sample, are particularly outstanding. Finally, it is noteworthy how the only city in the selection built by Le Corbusier shows a unique profile, different from the rest of the sample.

6. Conclusions

The first idea to note is the divergences found in terms of population data in reference to each of the cities designed by Le Corbusier. This was demonstrated in proposals such as Antwerp, which accounted for a population of 500,000 inhabitants; and others of a much smaller scale, such as the 5000 inhabitants in the case of Saint-Gaudens. This demonstrates the urban planner's interest in proposing the widest variety of urban solutions, as well as the diverse scale of cases he addressed.

In any case, the most interest in this section is that which is concerned with population densities since these are the data that allow a comparison between the proposals adopted for each of the cities independent of their size. Based on the density granted to residential areas, the plans can be classified into four groups. The study of the data coincides with the current climate in which advancements considering density levels of greater or lesser suitability for the post-COVID-19 city are being made (Córdoba Hernández et al., 2020:671).

Thus, in a high bracket, above 1000 inhab./ha, are Geneva, Antwerp and Berlin. In the next grouping, of approximately 500 inhab./ha, are Zlin and Bogota. The third grouping, between 150 and 300 inhab./ha, features Paris, Hellocourt and Meaux. Finally, with lower densities of fewer than 150 inhab./ha are Saint-Dié, Saint-Gaudens and Chandigarh. This last particular city, the only one to be built, has the lowest density of the entire sample while the proposal for Antwerp has the highest density. The variation in density between the two proposals is 1000%, that is, the density proposed for Antwerp is 10 times higher than that of Chandigarh.

It is also possible to identify general trends in the proposed densities based on the timeline analysis. It begins with Paris at a medium-low density, within the scale; and rises to approximately 1000 inhab./ha in the following two cities, figures that are not again repeated until the last project for the city of Berlin. The lowest values are in the central part, except for Bogota, which rises to approximately 500 inhab./ha. Thus, certain parallelism in stages is evident, so it is also possible to group proposals that are close to each other in time.

By measuring the surface areas allocated to each urban use, the set of cities studied was characterised in detail. The use of percentages made it possible to compare the different cases analysed independent of their scale. In this way, the four comparative linear graphs reveal the similarities and differences in the urban planning he undertook during his career.

Thus, in the analysis of how much surface area is allocated to residential use compared to other urban functions, there is a clear trend showing a balance in most of the plans of approximately 50%. This is particularly significant in the first cities examined in the study.

Subsequently, some variation was noted in cases that fell into a range of 40%—60% for either use. There are also two cases where this differentiation is more pronounced, namely, in Saint-Dié and Chandigarh, where residential use predominance is close to 80% of the total use.

In any case, it is necessary to note that this residential area is not composed only of housing. These projects, especially in Chandigarh, highlight a high mix of urban functions in residential areas. In the different cities, he proposes leisure and recreational uses alongside housing, a mixture that can even be found in Le Corbusier's buildings, such as the *Unités d'Habitation*.

However, the conception of the sectors would go further, being conceived as units based on self-sufficiency. In this way, the conception proposes to reduce displacements and promote social relations based on proximity. In this case, it comes from the 7 V model itself, where the city is configured through self-sufficient sectors. This mix of uses and self-sufficiency in proximity, together with the 15-min city, are presented as paradigms for improving the 21st century city, planning theories found in current urban approaches.

Among the various voices or examples found focusing on developing the city of proximity and commitment to pedestrian mobility, Paris stands out. It should be noted that Le Corbusier made different proposals for this French city, a place of constant current urban experimentation. Recently, the proposal called *"Le Paris du 1/4 heure"*, which advocates transforming the consolidated city so that routes are reduced as far as possible, was presented. This

project is based on research conducted at the Sorbonne and led by the urban planner Carlos Moreno (2020).

Concerning the quarter-hour city, Richard Sennett (2020) states the relevance of the review of architecture and urban density and the concentration of activities as challenges for urban planners today. Therefore, it is essential to characterise Le Corbusier's city and reanalyse his models to learn concepts for today's city.

Next, the proportion of built-up versus open space was reviewed and parameterised, a characteristic that is common throughout almost all of Le Corbusier's career as an urban planner. Here, except for Chandigarh, where more than 60% of the space is built-up versus 40% that is open space, all of the plans allocate more than 80% of the surface area to non-built-up space, either in the form of green spaces or pedestrian areas (most predominantly) or for roads. This demonstrates the importance Le Corbusier placed on open space as an essential characteristic of modern urban planning.

Once the areas not destined to be built upon had been defined, the urban surface allocated to pedestrians and vehicles was calculated. This analysis quantitatively illustrates the predominant recipient of his proposals according to the distribution of hectares.

In this case, the facts that all the cities allocate more space to pedestrians than to vehicles and that all of them obtain more than 60% of the total surface area available stand out. Within the entire sample, three cities stand out for close to 90% of the city designed for pedestrians: Antwerp, Saint-Dié and Berlin. The first two proposals, Paris and Geneva, and Chandigarh are still at over 60%.

Within the space allocated to green space and for recreation, there are proposals in which paths and walkways take centre stage. In some cases, they even take the form of a network that overlaps the urban areas.

Thus, although such intentions are only lightly outlined in the plans for Paris and are unable to be regarded as paths, they do appear in one form or another in every other one of the plans. The case of Chandigarh is particularly significant. In Chandigarh, not only is a network of pedestrian routes proposed, but larger open spaces also merge with them, changing the amount of space allocated to pedestrian movement in the process.

Finally, analysis of the amount of pedestrian area allocated to each inhabitant was conducted. The purpose of this analysis was to show how important this space was considered in each city. Thus, a large part of the sample shows between 14 and 45 m^2 allocated per inhabitant, with three cities standing out for having significantly higher amounts. These are Hellocourt and Saint-Dié, which exceeds 100 m^2/inhab.; and Saint-Gaudens, with a figure that no other city comes close to at approximately 200 m^2/inhab.

Given the results obtained from the analyses, the importance of recreational space for inhabitants in Corbusier's urbanism can be affirmed, in direct opposition to assertions that usually place the car at the centre of his conception of the city. This is not to say that he rejects vehicle movement but simply that it does not have the same prominence or carry the same weight when translated into surface area, as affirmed above.

The importance placed on the pedestrian at the expense of space for vehicles is something he alludes to in his texts: "We have given all the soil of the city to the pedestrian, on the land itself: lawns, trees, playgrounds: about 100% of the soil is for the use of the inhabitants [...] I would add: a pedestrian never meets a car, never" (Le Corbusier, 1964: 108). This point is demonstrated in the current research and vindicates the rereading of Le Corbusier's own texts as a means to overcome preconceived ideas about his urban planning or about the drift that eventually led to the modern city.

Alongside the validity of the study in terms of density for a post-COVID-19 city, there is a parallelism in terms of the reflection on and the need for open spaces with a greater allocation to pedestrians than to vehicles (Córdoba Hernández et al., 2020: 666). What is particularly relevant is the reconversion of roadways into pedestrianized areas, mainly at historic sites (Mardones-Fernández de Valderrama et al., 2020: 656—657). For this reason, it is crucial to review Corbusian urbanism in which pedestrian areas take precedence over roadways. This could shed light on the urban planning of the post-pandemic twenty-first-century city.

Finally, the review of all the inland cities together and comparatively shows the diversity of Le Corbusier's work regarding urbanism. Different models are adopted and vary throughout the course of his career, and they are modified and adapted depending on the size and context of the city in question. With all this in mind, there is advocation to venture beyond Le Corbusier's urbanism, as seen through the lens that is the Plan Voisin and the Athens Charter.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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